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Effect of Graft copolymerization of Oleic Acid on to Cyclic Natural Rubber in Polyimide

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ABSTRACT

In this work, graft co-polymerization of cyclic natural rubber (CNR) with oleic acid (OA) using dicumyl peroxide (DCP) as initiator was investigated, owing to the low-cost and eco-friendly materials of OA. The grafted CNR with OA was characterized using FTIR, DSC, percentage of grafting and SEM. Carbonyl group chain (C=O) present in CNR when adding 9 parts per hundred rubber (phr) of OA, thus confirming the grafted reaction process and resulting in percentage of grafting of 3.2 %. Thermal characterization based on DSC suggesting that the shifted of Tg to higher temperature with incorporation of 9 phr of OA, which can be ascribed to the carbonyl group was grafted onto polymer frame of CNR cause free-rotational become reduce, thus higher Tg for CNR-g-OA was achieved. Furthermore, morphology analysis demonstrated that the interfacial adhesion between PI and CNR was increase with the present of 9 % CNR-g-OA as indicated by homogenous, uniform dispersion and smooth surface. This

finding suggests that CNR-g-OA suitable as compatibilizer for paint industry application.

Keywords: Cyclic natural rubber, oleic acid, graft, thermal, polyimide

Introduction

Resiprene-35 or cyclic natural rubber (CNR) are chemical modification of natural rubber through cyclization of intermolecular cis-1,4-polyisopropene with lewis acid compound [1-4]. The application of CNR are commonly used as paint binder, printed ink and adhesive agent due to its advantages of chemical resistance and water and good adhesive properties on metal, glass, wood and paper. However, CNR suffered from physicochemical properties including its compatibility on polar surface and adhesive force of CNR as paint binder are weak towards polymer with polar properties such as polyimide (PI) [5-8].

In general, CNR are natural binding agent for paint since its non-polar, however weak surface adhesion and prone to the free-radical such as anorganic acid and ozone due to double bond ($C=C$) on cyclic carbon chain, meanwhile PI is commonly used as synthetic paint with polar properties, high surface adhesive, abrasion resistance, high pH and high temperature [9]. Consequently, when PI and CNR was mixed together formed a non-polar system without chemical bond and immiscible, therefore resulting in poor mechanical properties [10-16]. Main reason behind the immiscible of PI and CNR are the incompatible between two materials, since low interaction between PI-CNR. In order to increase the compatibility of PA-CNR, it required an compatibilizer to obtain miscible and homogenous mixture [7,8,15-23]. Extensive research have been conducted in order to improve the compatibility by modifying CNR with maleic

anhydride monomer through graft copolymerization using benzoyl peroxide [5]. Previous work has been successfully grafted OA into acrylonitrile butadiene styrene (ABS) using BPO initiator namely ABS-g-OA using reflux methods [24]. In addition, glycidyl methacrylate (GMA) was grafted onto NR (NR-g-GMA) to improve compatibility and toughness of NR/polylactic acid (PLA) blend [25]. Borapak *et al.* modified NR grafted with poly(2-hydroxyethyl acrylate) to increase polarity to compatibilize NR/silica blend [26]. Recent suggest that graft oleic acid are compatible with linear low density polyethelene [27] graphene [28–30] and chitosan [31] for antifog, corrosion protection and chemosensor, respectively.

In this work, OA was utilized not only to modify the physical and mechanical properties of CNR, but also improving the compatibility, since OA can be converted into epoxide oil for grafting monomer, thus the grafting OA on CNR could produce CNR-g-OA. Our finding demonstrated that incorporating compatibilizer on homogenous mixture PI-CNR can be achieved as binder agent for paint.

Materials and methods

Cyclic natural rubber (resiprene-35; $\rho = 0.91$ g/mL, base ≤ 5 mg KOH/g) was obtained from PT. Industri Karet Nusantara, Deli Serdang-North Sumatera, Indonesia. Polyimide, dicumyl peroxide (99 %), oleic acid (98 %), xylene (99.8 %) and KOH (99%) was purchased from Merck. Internal mixer Thermo haake polydrive (Brabender Plasticoder PLE 331-W50H, Germany).

Preparation of CNR-g-OA

30 g of CNR was added into internal mixer at 165 °C for 8 min with rotor speed of 80 rpm. After CNR melt, 9 phr of OA, 1.5 phr of both dicumyl peroxide (DCP) and divinylbenzene was added into above mixture for 4 min (grafting reaction), then the

warm CNR was kept for 24 h at room temperature. The composition of OA monomer and DCP are shown in Table 1.

Table 1. Various composition of OA and DCP.

Sample	CNR		OA		DCP	
	(phr)	(g)	(phr)	(g)	(phr)	(g)
OA3			3	0.9	1.5	0.45
OA6	100	30	6	1.8	1.5	0.45
OA9			9	2.7	1.5	0.45
OA12			12	3.6	1.5	0.45

(phr = parts per hundred rubber)

The mixture of PI/CNR was prepared with internal mixer without and with compatibilizer of CNR-g-OA around 30 g (Table 2) was mixed homogenously in internal mixture at 170 °C for 8 min with rotor speed of 80 rpm.

Table 2. Various mixtures composition of CNR with and without CNR-g-OA.

Sample	PI	CNR	Compatibilizer CNR-g-
	(% b/b)	(% b/b)	OA
			(% b/b)
PA/CNR20	80	20	-
PA/CNR30	70	30	-
PA/CNR40	60	40	-
	80	20	4

PA/CNR20/CNR-	80	20	6
g-OA	80	20	8

Percentage of Grafting via titration method

In order to determine the percentage grafting of CNR for without and without OA, the titration method was employed. 1 g of CNR-g-OA was added into Erlenmeyer flask and dissolved in 50 mL of xylene, then heated until it melt, afterwards 3-4 drops of phenolphthalein (PP) was added with warm temperature of titration (KOH-ethanol solution) until the solution turn into faded red color. The percentage of grafting was calculated using the following equation:

$$(\%) \text{ Grafting} = \frac{N [(V_o - V)] M}{W \times 1000} \times 100\%$$

where, N represent concentration of KOH (0.05 N), V represent the titration volume, V_o correspond to the empty volume, M correspond to molecular weight of OA and W represent the weight of sample.

Characterization

Characterization was carried out using fourier-transform infrared spectroscopy (FTIR) Prestige Shimadzu 8210PC, differential scanning calorimetry (DSC) Shimadzu 2910, and scanning electron microscope (SEM) Hitachi TM4000.

Results and discussion

Figure 1 shows the without and with CNR grafted at various concentration of 3, 6, 9 and 12 phr of OA. The characteristics of CNR spectrum showed at 1605 cm^{-1} corresponds to

the C=C stretching, and 748 cm^{-1} can be ascribe to the C-H bending vibration [1,2,23,33].

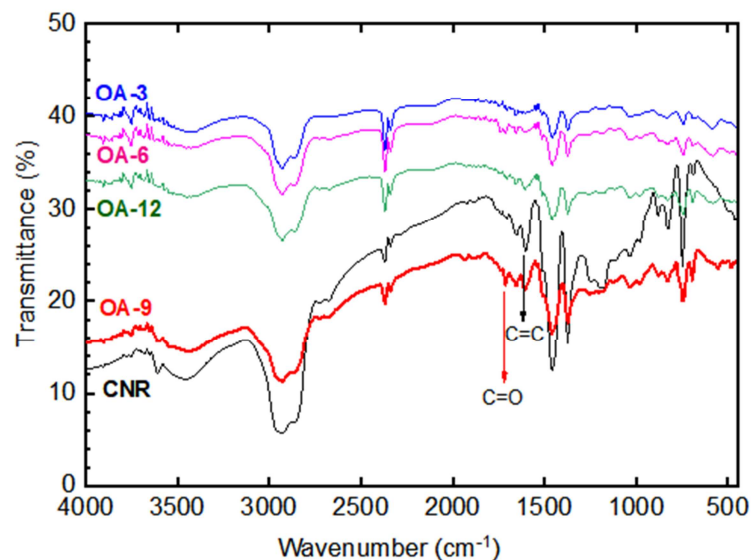


Figure 1. FTIR spectra of CNR-g-OA with various concentration of OA.

OA-3 phr, OA-6 phr, and OA-12 phr showed the presence of OH stretching from the carboxylic group at peaks 3449 cm^{-1} [34,35] and C-H vibrational stretching at peaks 2932 cm^{-1} correspond to the CNR molecule [36,37]. In the fingerprint region between 1500 cm^{-1} to 500 cm^{-1} , the peak at 1458 cm^{-1} correspond to the $-\text{CH}_2$ stretching, and at 1373 cm^{-1} correlated to a $-\text{CH}_3$ group. In the CNR polymer chain, the C-C stretching observed at 1119 cm^{-1} [2,32,33]. A sharp absorption peaks with low intensity was observed at 1701 cm^{-1} which correspond to the presence of C=O groups only for OA-9 sample. This finding indicates that OA has been grafted on CNR even though the intensity is relatively low [24,38].

The effect of OA on CNR-g-OA grafting percentage was evaluated by titration methods. The variation of OA monomer concentration and the calculated PG are presented in Table 3.

Table 3. The percentage of grafting CNR-g-OA with variation of OA concentration.

Sample	CNR	OA	DCP	Titration	Percentage of
	(phr)	(phr)	(phr)	Volume (mL)	grafting
					(%)
		3	1.5	5.4	2.50
CNR-g-	100	6	1.5	5.2	2.79
OA		9	1.5	4.9	3.20
		12	1.5	5.1	2.93

The percentage of grafting for 3, 6, and 9 phr of OA gradually increased as depicted in Figure 2. It caused the propagation rate of radical OA monomer chains to become slow, affected the OA homo-polymer chain on CNR were shortened. Accordingly, OA-9 phr showed the optimum concentration with the highest percentage of grafting 3.20%, meanwhile at higher concentration of OA up to 12 phr resulting in decrease of grafting percentage. This finding implying that the formation of long radical OA monomer chains cause decrement of percentage of grafting, which may be due to OA was inhibited to join the active site of the CNR due to the slow movement of radical homo-polymer. This trend was in accordance with previous report [24,38,39,40-43].

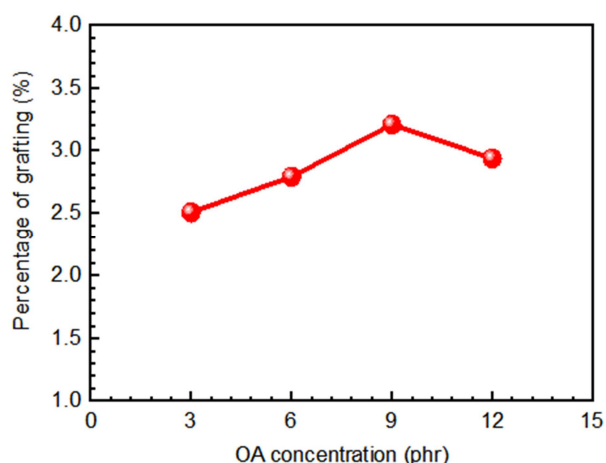


Figure 2. Percentage of grafting as a function of OA concentration.

The effect of DCP initiators on CNR-g-OA grafting percentage was evaluated by titration methods. Mass ratio CNR: OA remained constant while the varied concentration of DCP 0.5, 1.0, and 1.5 phr was utilized as shown in Table 4.

Table 4. Percentage of grafting CNR-g-OA with the variation of DCP.

Sample	CNR (phr)	OA (phr)	DCP (phr)	Mass (g)	Titration Volume (mL)	Percentage of grafting (%)
		9	0.5	1.0152	5.2	2.78
CNR-g-	100	9	1.0	1.0134	5.1	2.92
OA		9	1.5	1.0123	4.8	3.34

Table 4 showed that the grafting percentage gradually increased by the increase of DCP. The highest percentage of grafting was 3.34% for DCP of 1.5 phr and the lowest was 2.78 % for DCP of 0.5 phr. The DCP initiator play a significant role in the OA grafting reaction process onto CNR molecular chain to produce CNR-g-OA. The free radical concentration increased during the increasing of DCP concentration. It benefited the grafting copolymerization that increased the percentage of grafting [38].

Thermal Analysis

In order to understand the glass transition temperature (T_g) and melting temperature of graft copolymerization process of OA and CNR before and after grafted, DSC was carried out. Figure 4a and b shows the DSC curves of before and after CNR grafted. The blue and red line indicates the heating process from 30-150 °C with interval of 10 °C/min for first and second thermal sweep line, followed by cooling process to room temperature.

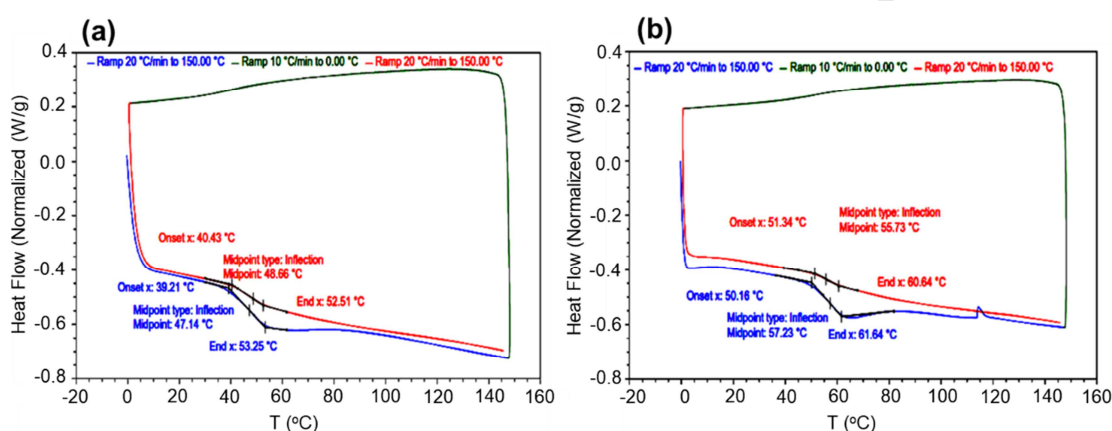


Figure 3. DSC curves of (a) before and (b) after grafted 9 % CNR-g-OA.

The T_g and T_m value was determined from second thermal scan [44]. Based on Figure 3a and b, pristine CNR demonstrated T_g of 48.66 °C before grafting process and shifted to higher temperature of 55.73 °C. Similarly, the T_m of CNR before and after OA graft showed slightly shifted to higher temperature from 146.5 to 147.50 °C. The increased of T_g was in accordance with additional OA monomer. More OA group grafted onto CNR resulting in more rigid and polar surface, which indicates the substitution of carbonyl group of OA (Figure 1) with polymer frame of CNR, hence the free-rotational becomes reduce and higher T_g was obtained [24,38,45,46]. Decrease in T_g and T_m also implying the decrease in amorphous in natural rubber which leads to enhancement of flexibility and softening of the polymer backbone [47]

Morphology analysis

In order to investigate the homogenous of PI/CNR and its compatibility without and with CNR-g-OA, SEM was employed. Figure 4a exhibits the surface morphology of PI/CNR (80:20) without and with additional CNR-g-OA at various concentration. The mixture between PI and CNR without compatibilizer resulting in two different phases (red line) as shown in Figure 4a indicated that the surface are not uniformly distributed and incompatible due to large grain-like and rough (large pore). CNR are mainly not disperse with PI matrix due to the poor adhesion properties between two polymers, thus poor mechanical properties. This finding is in good agreement with the immiscible of PI and CNR due to the polarity and adhesion properties, thus resulting in poor mechanical property [48,49].

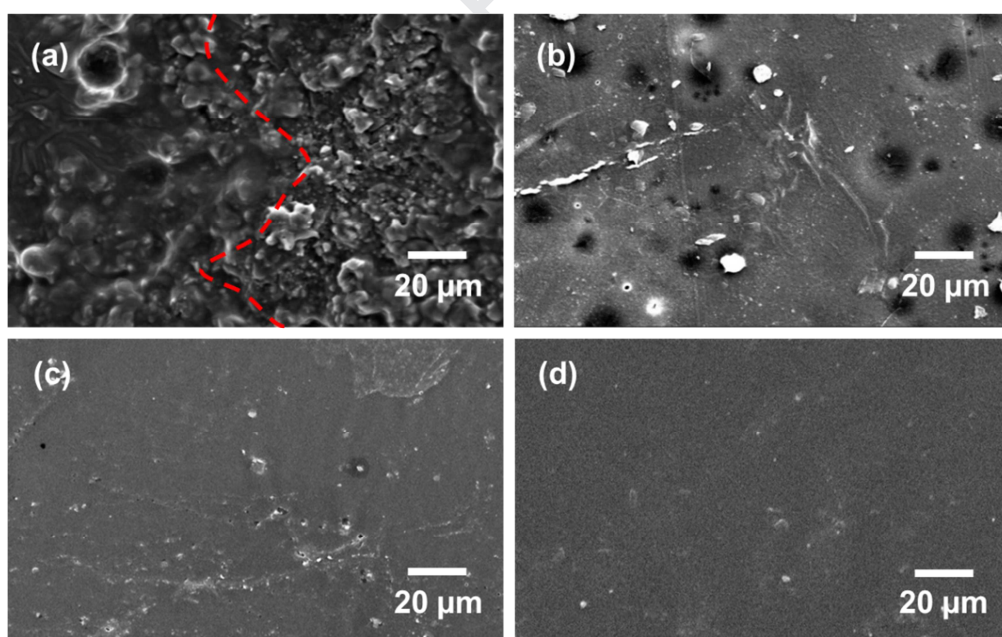


Figure 4. SEM images of (a) pristine PI/CNR, (b) 3, (c) 6, and (d) 9 % of CNR-g-OA incorporated into PI/CNR.

Figure 4b shows the PI/CNR after incorporated with 4% of CNR-g-OA with relatively smoother surface (only smaller particle present) as compared to pristine

PI/CNR, even though the formation of small pore-like are still appeared. Additional 6 % of CNR-g-OA displayed homogenous and better compatibility of PI/CNR indicates the adhesion property of PI/CNR slightly improved and not only reducing the brittleness of pristine CNR but also CNR-g-OA act as efficient bridge between physical interaction of PI and CNR. This finding is supported by previous report [8,12,14,15]. Further increasing the concentration of CNR-g-OA up to 8 % demonstrated that the surface morphology of PI/CNR are smoother, high compatibility, where particles are uniformly distributed and absence of pore formation. This finding indicates that surface adhesion increased, therefore resulting in good mechanical property.

Conclusion

In summary, graft copolymerization of OA onto CNR using DCP initiator was successfully prepared with different concentration from 3 to 9 phr OA into the CNR/PI blend matrix in order to examine the chemical, percentage of grafting, thermal and morphology properties. Graft copolymerization of OA onto CNR was successfully prepared based on the percentage of grafting, FTIR and DSC analysis. The increment of percentage of grafting correlated with increased of concentration up to 9 phr of OA and DCP initiator of 1.5 phr, which cause the formation of free radical from DCP increases. Based on FTIR analysis, the presence of carbonyl group (C=O) from OA confirming that CNR was grafted. In addition, thermal analysis using DSC indicates the T_g shifted from 48.66 to 55.73 °C with additional OA monomer during grafting process. Accordingly, the morphology of PI/CNR incorporated with 9 % CNR-g-OA was significantly improved due to increase of interfacial adhesion, better dispersion and absence of pore formation.

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Author Statement

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Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: